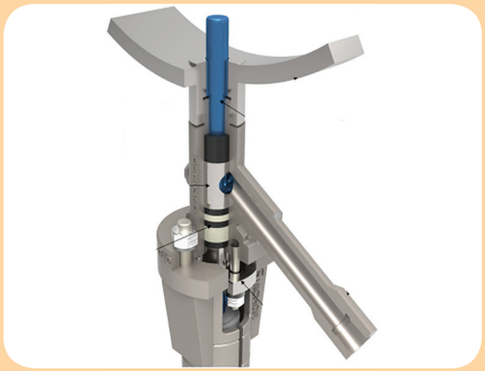
What is a Piston Valve?

- Piston valve was developed in 19th century.
- It is a device used to control the motion of a fluid along a tube or pipe by means of the linear motion of a piston.
- They are ideally recommended for critical and hazardous media, including Steam, Heat transfer oils, acids and gases.



like products when used. Hence they are not generally used in motor vehicles.

8.3.3.2 Vegetable Oils

They are obtained from vegetable seeds. For example: Linseed oil, castor oil and palm oil. Except castor oil all the vegetable oils easily get converted into gum like materials. At high temperatures viscosity of the castor oil decreases. Castor oils were used in old vehicles. However it is not used in modern vehicles.

8.3.3.3 Mineral Oils

These are the oils mostly used in all vehicles. These lubricating oils are obtained as lubricants in the separating processing of petroleum refining. Their important properties do not vary significantly even at high temperatures. They do not dissolve in water, acid free and corrosion free. Hence they are largely used in all automobiles.

8.4 TYPES OF LUBRICATION SYSTEM

All the parts of engine such as crankshaft, bearings, crank pin, both ends of connecting rod, piston pin and inner wall of cylinder, piston rings, valve mechanism, timing gears and camshaft bearings are made to function along with the engine. Hence the above parts must be lubricated to avoid friction in these moving parts. The following are the types of lubricating systems used in engines.

Lubricant Systems:

1. Petrol oil (mist) lubricating system.
2. Splash lubricating system.
3. Pressure lubricating system.
4. Semi pressure lubricating system.

- Oil sump
- Oil pump
- Pick-up screen
- Pressure regulator
- Oil filter
- By-pass valve
- Oil galleries
- Dipstick
- Pressure indicator

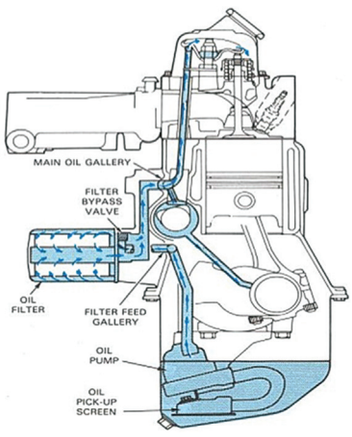


Figure 8.4 Parts of Lubrication System

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